

Plotting, Ordering and Comparing Real NumbersCircuit 7A.02.06 **ANSWER KEY**

Work the four parts in order, on your own. Everything here is from Lessons 1 through 5. This is the last practice before the unit assessment.

PART 1. Comparing

Use $<$, $>$ or $=$. One of the six is an equality.

1.	-5π	$<$	$-\sqrt{225}$
2.	$\sqrt[3]{125}$	$=$	$\frac{\sqrt{100}}{2}$
3.	$\sqrt{45}$	$<$	$\pi + 4$
4.	-7	$<$	$-\sqrt[3]{49}$
5.	$\frac{29}{3}$	$>$	$\sqrt{80}$
6.	$\frac{\pi}{3}$	$<$	$\sqrt{22} - 3$

Values: -15.708 vs -15 ; 5 vs 5 ; 6.708 vs 7.142 ; -7 vs -3.659 ; 9.667 vs 8.944 ; 1.047 vs 1.690 . Row 4 is the one to watch — $-\sqrt[3]{49}$ is about -3.7 , which is well to the right of -7 . Row 2 is the equality and a student who never writes $=$ has not checked it.

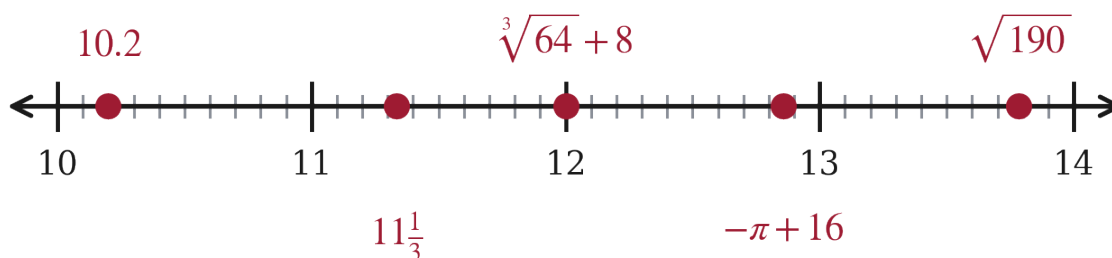
PART 2. Sorting and plotting

A number is irrational only if it cannot be written as a ratio of two integers.

7. Sort the five expressions, then plot and label each one on the number line.

$$10.2 \quad 11\frac{1}{3} \quad \sqrt[3]{64} + 8 \quad -\pi + 16 \quad \sqrt{190}$$

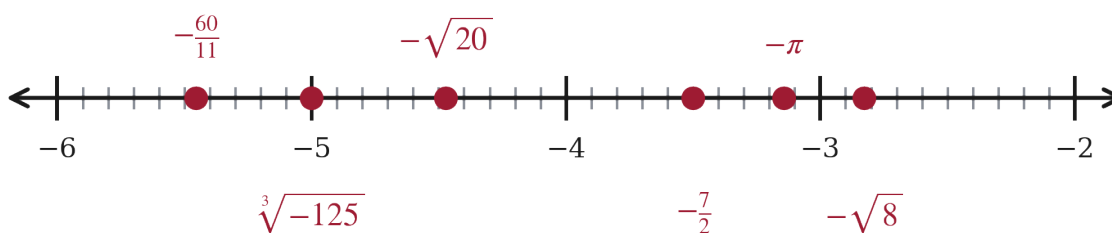
Only rational numbers	Includes an irrational number
10.2	$-\pi + 16$
$11\frac{1}{3}$	$\sqrt{190}$
$\sqrt[3]{64} + 8$	



$\sqrt[3]{64} + 8 = 4 + 8 = 12$ exactly, so it is rational. Values: 10.2; 11.333; 12; 12.858; 13.784.

8. Plot and label each expression on the number line.

$$-\frac{60}{11} \quad \sqrt[3]{-125} \quad -\sqrt{20} \quad -\frac{7}{2} \quad -\pi \quad -\sqrt{8}$$



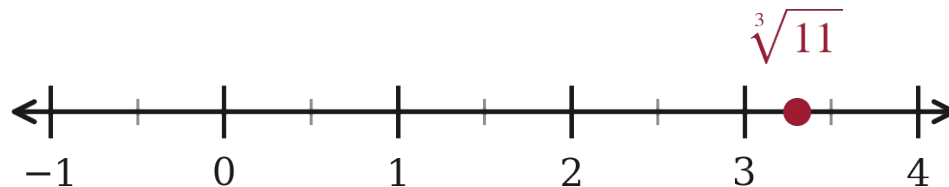
Values: -5.455 ; -5 ; -4.472 ; -3.5 ; -3.142 ; -2.828 . All six are negative, so the one that looks biggest is furthest left.

PART 3. Error analysis

Read the work, find the error, then do it correctly.

9. Sara approximated $\sqrt[3]{11}$ like this:

“11 is between 9 and 16, which are both perfect squares. So $\sqrt[3]{11}$ lies between 3 and 4, and it is closer to 3 because 11 is closer to 9 than to 16.”



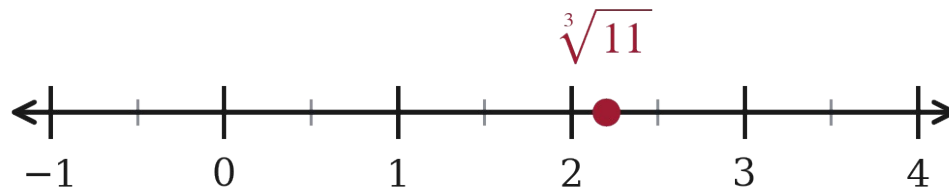
- a. What is Sara's error?

She used perfect squares to estimate a cube root. She should have used perfect cubes.

- b. Redo it correctly, then plot the right value.

11 lies between the perfect cubes 8 and 27, so $\sqrt[3]{11}$ lies between 2 and 3.

11 - 8 = 3 and 27 - 11 = 16, so 11 is much closer to 8. $\sqrt[3]{11} \approx 2.2$



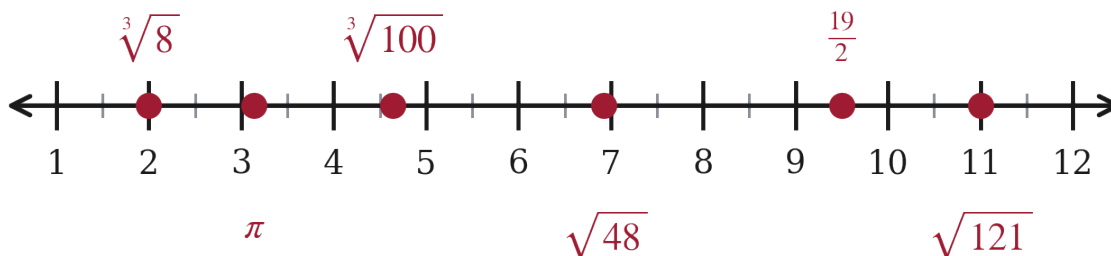
Sara's method is right and her list is wrong. Say that out loud — she is one word away from a correct answer, and the word is “cubes.”

PART 4. Ordering and plotting

10. Order the numbers from least to greatest, then plot and label each one.

$$\sqrt[3]{8} \quad \pi \quad \sqrt[3]{100} \quad \sqrt{48} \quad \frac{19}{2} \quad \sqrt{121}$$

$$\sqrt[3]{8}, \pi, \sqrt[3]{100}, \sqrt{48}, 19/2, \sqrt{121}$$

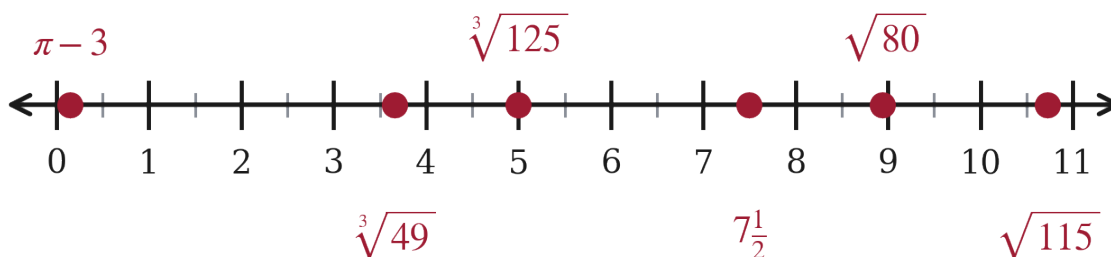


Values: 2; 3.142; 4.642; 6.928; 9.5; 11. Two of the six are whole numbers in disguise.

11. Order the numbers from least to greatest, then plot and label each one.

$$\pi - 3 \quad \sqrt[3]{49} \quad \sqrt[3]{125} \quad 7\frac{1}{2} \quad \sqrt{80} \quad \sqrt{115}$$

$$\pi - 3, \sqrt[3]{49}, \sqrt[3]{125}, 7\frac{1}{2}, \sqrt{80}, \sqrt{115}$$



Values: 0.142; 3.659; 5; 7.5; 8.944; 10.724.

STRETCH. Beyond the range the test uses. Same method, bigger cubes.

12. Find the value of $\sqrt[3]{216}$ and of $\sqrt[3]{-343}$.

$$\sqrt[3]{216} = 6, \text{ because } 6 \times 6 \times 6 = 216. \quad \sqrt[3]{-343} = -7, \text{ because } (-7)^3 = -343.$$